WRMatch: Improving FixMatch With Weighted Nuclear-Norm Regularization for Few-Shot Remote Sensing Scene Classification

Yunsheng Xiong, Kele Xu, *Member, IEEE*, Yong Dou, Yang Zhao, and Zikai Gao

***Abstract*— Semisupervised learning (SSL), such as FixMatch, has been successfully applied to remote sensing scene classification to relieve the burden of data annotation. However, in some extreme settings, only very few samples available, e.g., one to ten labels per remote sensing scene, can be used. When meeting this "few-shot" scenario, the deep model may be overfitting and prone to generate confusing predictions due to the lack of labels and strong augmentation-based perturbations. Thus, the prediction's diversity may collapse, and the discriminability exceeds the reasonable interval. How to improve the performance of few-shot learning is underexplored for the remote sensing scene classification in previous studies. In this article, we present a novel framework for the task by utilizing the improved FixMatch and the weighted nuclear-norm regularization (WNNR). Specifically, we regularize the prediction matrix by exploiting the nuclear-norm, which is an approximation of the matrix rank and a relaxed boundary for the Shannon entropy. We further provide two weighting schemes to improve the nuclear-norm-based regularization. First, the random-weighting scheme for nuclear-norm (RWNNR) is proposed based on the Dirichlet distribution to improve the model's generalization. Second, we present the self-weighting scheme (SWNNR) to weight the singular values according to singular values themselves and adjust the relaxed degree for the boundary between the nuclear-norm and the Shannon entropy. Maximizing the weighted nuclear-norm can improve the prediction diversity and optimize the prediction discriminability simultaneously. Combining the advantages of SSL and the aforementioned improvements, we can reliably classify the remote sensing scene image with very limited annotated datasets. To empirically demonstrate the proposed method's effectiveness, we comprehensively evaluate the method on three publicly available benchmark datasets. The results show that the proposed method outperforms the baseline methods by a large margin and achieves superior performance on all three datasets. Our method can be an effective alternative to metalearning in few-shot scene classification, with the advantage lying in the competitive performance and the absence of metatraining stage associated with a large number of labels.**

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***Index Terms*— Discriminability, diversity, few-shot, remote sensing, scene classification, semisupervised learning (SSL), weighted nuclear-norm.**

I. Introduction

With the rapid development of space science and technology, remote sensing image plays an increasingly critical role in both military and civil fields [1]–[3]. As an essential component in the interpretation of remote sensing imagery, the scene classification task is highly related to other tasks, such as target detection, image retrieval, and anomaly detection [3]. It is desirable to design an automatic scene classification method with high robustness and good generalization ability, as its applications seem to be evident in several fields.

In the last decades, sustainable efforts have been made, and deep learning has become the methodology of choice for remote sensing scene classification [4]. Convolutional neural networks (CNNs) were widely applied for the classification task. Due to the excellent feature representation capability, CNNs have prevailed in the area for remote sensing scene classification [5]–[8]. Generally, CNNs require a massive number of parameters and large amounts of labeled examples to train the neural network with good generalization performance and prevent overfitting [9]. However, the annotation of remote sensing data is complicated and requires strong domain expertise; sometimes, even auxiliary means are needed. Moreover, compared to natural images, remote sensing images are taken from a remote vantage point in space. They often cover a much wider surface, have fewer subtle features, and contain far more complex object distributions. As shown in Fig. 1, the visual characteristics of objects in remote sensing images are quite different from the visual contents that are seen in natural images. The annotation is a time-consuming and laborious job, which hinders the large-scale application of deep learning in remote sensing image processing. On the other hand, sometimes, we have to deal with a scenario with extremely limited labels, e.g., one category may have fewer than ten labels. Due to the high interclass similarity in remote sensing scene images (as shown in Fig. 1), the model will generate confused predictions.

Few-shot learning can greatly alleviate the burden of data annotation and can be utilized for the scene classification task. For the few-shot scene classification, metalearning [1], [2] is

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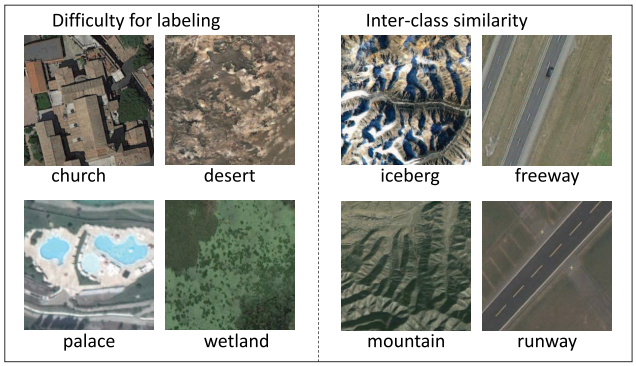


Fig. 1. **Examples of remote sensing scene images, which are used to illustrate the motivations of this article.** (Left) Remote sensing images have unique perspectives, complex compositions, and lacking details, which makes manual labeling very difficult. (Right) Due to the high interclass similarity in remote sensing scene images, the model can easily generate confusing predictions when labels are extremely scarce. Consequently, a special mechanism is needed to ensure the diversity and discriminability of prediction in these extreme settings.

the mainstream method and achieves state-of-the-art performance. Metalearning includes two phases: metatraining and metatesting. In the metatraining stage, a series of metatasks is constructed to train the model, and each task has its own labeled training set (support set) and test set (query set). In the whole training process, different metatasks will be sampled in each training, and each metatask contains different combinations of categories. This mechanism enables the model to learn the common parts of different metatasks. In the metatesting stage, the model can also classify them well when facing new and unprecedented metatasks. Although each metatask needs only a very small number of labels, the metatraining phase composed of multiple metatasks requires a large number of labels.

Semisupervised learning (SSL) is another effective way to reduce the requirement for labeling, which aims to leverage large amounts of unlabeled data to improve the performance of supervised learning over small datasets [10], [11]. In the remote sensing analysis fields, many efforts have been made to develop semisupervised methods, but few works utilize it to address the "few-shot" case. A semisupervised approach would eliminate the metatraining phase of metalearning and the large number of labels associated with that phase. As the state-of-the-art SSL algorithm, FixMatch combines the augmentation-based consistency regularization (CR) and pseudolabeling in a unified framework and can train in an end-to-end manner. When directly applying FixMatch to few-shot scene classification, however, we can observe the model's diversity collapses, and an overfitting phenomenon occurs. In this article, we propose an improving semisupervised algorithm, named WRMatch, for few-shot remote sensing scene classification. Leveraging the nuclear-norm of the model's output matrix to enlarge prediction diversity, we provide two schemes to weight each singular value. By maximizing the weighted nuclear-norm, our method can improve the prediction diversity and adjust the discriminability in a reasonable range. In short, the main contributions of this article can be summarized as follows.

1) We propose a novel framework for the remote-sensing scene classification task within the few-shot settings. Unlike many of the previous few-shot learning methods (especially the metalearning-based) that require a large number of labels in the metatraining phase and can only alleviate the need for labels of the unseen category during the metatest phase, we explore the SSL approach, such as FixMatch, for the classification task. Our approach can greatly reduce the annotation demand for the training.
2) When directly applying FixMatch to few-shot scene classification, we find the model's diversity may collapse, and an overfitting phenomenon occurs. We conduct an in-depth analysis of the phenomenon and investigate the intrinsic correlation among nuclear-norm, diversity, and discriminability.
3) To improve the prediction diversity, we exploit nuclear-norm to regularize the model's output matrix. To avoid overconfident predictions, we provide two schemes to adjust the prediction discriminability: the random-weighting scheme for nuclear-norm (RWNNR) based on the Dirichlet distribution and the self-weighting scheme (SWNNR) to weight the singular values according to singular values themselves. Therefore, our method can improve the diversity and optimize the discriminability of prediction simultaneously.
4) The empirical experiments show that the proposed method outperforms the existing baseline methods by a large margin and achieves superior results on all three datasets. Our method can be an effective alternative to metalearning for the few-shot remote sensing scene classification, which could produce competitive classification accuracy and eliminate the metatraining phase of metalearning and the large number of labels associated with the learning phase.

The remaining of this article is organized as follows. Section II discusses the relationship between the proposed method and prior works. Section III provides the details of our methodology. Section IV describes comprehensive experimental results. Section V discusses the strengths and limitations of our method. Section VI provides the conclusion of this article.

II. Related Works

In this section, we will review the related works for remote sensing scene classification. As our method is related to few-shot learning and SSL, we also discuss both.

A. Remote Sensing Scene Classification

Recently deep learning has dominated the field of remote sensing scene classification [5], [6], [12]. Marmanis *et al.* [6] proposed a two-stage method that exploits a pretrained CNN to extract initial representations, and the derived representations are then transferred into a supervised classifier. Muhammad *et al.* [13] employed the pretrained VGGNet as a deep feature extractor to extract mid-level and deep features. Chen *et al.* [14] proposed a CNN model under the guidance of a human visual attention mechanism. Li *et al.* [15] proposed

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an ensemble model with a cascade attention mechanism. Li *et al.* [16] proposed an attention mechanism-based CNN with multiaugmented schemes. Alharbi *et al.* [17] presented a novel method for selective data augmentation in remote sensing and tried to optimize the way we augment the training set. To alleviate data annotation, most of CNNs take advantage of transfer learning (e.g., the pretrained models in ImageNet [18]) and data augmentation. Although these methods can significantly reduce the demand for hand labeling work, a considerable number of labels are still necessary to achieve the desired performance.

B. Few-Shot Learning

Few-shot learning is a special case of machine learning, which implements target tasks (such as classification and detection) with only a small amount of labels provided by the training set. Data augmentation and data synthesis (e.g., generative adversarial networks [19]) provide basic means of adding labeled examples, but such methods are not sufficient to effectively improve the generalization capacity of the model. Transfer learning [20], [21] exploits other datasets and could provide extra information. However, differences between source and target datasets can seriously degrade the performance of the transferred model. For few-shot scene classification in remote sensing, most advanced methods are based on metalearning [1], [2], [22]. Many classical metalearning algorithms, such as Matching Nets [23], prototypical networks [24], and model agnostic metalearning [25], have been successfully applied to this task. Yuan *et al.* [26] combined multiple attention mechanisms and the attention-reference mechanism into the deepEMD network in a metalearning manner. Li *et al.* [2] proposed an end-to-end network for boosting a few-shot remote sensing scene classification, called discriminative learning of adaptive match network (DLA-MatchNet). Li *et al.* [1] proposed a simple but effective method, called RS-MetaNet, to resolve the issues related to few-shot remote sensing scene classification in the real world. The downside of metalearning is that it requires a metatraining phase, which requires large amounts of labels for training. For example, in the experiment of DLA-MatchNet, the RESISC45 [27] dataset is split into 25, ten, and ten classes for training, validation, and testing, and there are 700 examples every class, which means that 17 500 labels are needed for metatraining. Only after extensive training, a model can be fine-tuned with a small number of samples to recognize new and unseen categories.

C. Semisupervised Learning

SSL is a combination of supervised learning with unsupervised learning, which employs large amounts of unlabeled data with a small amount of labeled data at the same time for pattern recognition. Many approaches have been explored for SSL, including handcrafted feature-based methods [28], [29], graph-based methods [30]–[32], pseudolabeling-based methods [33], [34], CR-based methods [35], [36], and so on. Self-training [33] and cotraining [34] are commonly used methods in remote sensing image processing to obtain pseudolabels. Aydav and Minz [37] utilized granular computing concepts to improve the self-training approach for the classification of the remote sensing images. Zhou *et al.* [34] presented a method based on semisupervised stacked autoencoders (Semi-SAEs) in a semisupervised cotraining fashion. Ma *et al.* [38] proposed a SiftingGAN approach to generate more diverse, more authentic labeled samples for data augmentation. Zhang *et al.* [39] proposed a novel noisy label distillation (NLD) method based on the teacher–student framework. Wang *et al.* [40] first designed a CR training method for semisupervised training. Li *et al.* [35] introduced the virtual adversarial training (VAT) method that maintains consistency before and after adding adversarial perturbation. Wang and Wang [36] proposed the VAT, aimed to regularize the network. Recently, a hybrid approach called FixMatch [41] combines pseudolabeling and CR to achieve SOTA performance. Besides, some works have attempted to combine semisupervised methods with few-shot learning. Boney and Ilin [42] and Ren *et al.* [43] utilized unlabeled data to improve the prototypical networks [24] and elevate the generalization ability of the model. Ye *et al.* [44] utilized unlabeled data from a macro perspective to form unsupervised few-shot tasks and encouraged close tasks to behave similarly. The downside is that they are still based on the basic process of metalearning and need a large number of tasks. Although SSL greatly reduces the burden of data annotation, few works have been made to extend SSL to few-shot scene classification in remote sensing. In this article, we strive to promote this work and to address the newly arising problems in this scenario.

III. Methodology

In this work, we target the remote sensing scene image classification task leveraging SSL with very limited labeled data. The main idea of the proposed method is depicted in Fig. 2. When applying FixMatch to the few-shot scene classification, there appears obvious diversity collapse and overfitting phenomenon. We analyze the intrinsic causes and improve the prediction diversity by maximizing the nuclear-norm of the output matrix. To keep the prediction discriminability in a reasonable range, we propose two weighted schemes for nuclear-norm, which can further improve the classification accuracy. We will explain the method in more detail subsequently.

A. Background

To illustrate the basic principles of SSL, we first provide the related background and notations. Let $X = \{(x_i): i \in (1, \ldots, N)\}$ refer to N examples in a batch, while $Y = \{(y_i): i \in (1, \ldots, N)\}$ is a batch of one-hot labels. $L = \{(x_i, y_i): i \in (1, \ldots, N)\}$ consists of N labeled examples, where x_i are the training examples and y_i are one-hot labels. $U = \{u_i: si \in (1, \ldots, N)\}$ is a batch of unlabeled examples, while $\tilde{U} = \{\tilde{u}_i: i \in (1, \ldots, N)\}$ refers to N unlabeled augmented examples in a batch, where $\tilde{u}_i$ is used to denote the unlabeled example that perturbed by augmentations. $P_\theta(y'|x)$ is the prediction generated by the model for input x, while $P_\theta(Y'|X) = \{P_\theta(y'_i|x_i): i \in (1, \ldots, N)\}$ presents a batch of predictions; here, θ refers to the parameters of the model. $F = \{(f_i): i \in (1, \ldots, N)\}$ consists of N pseudolabels, where f_i

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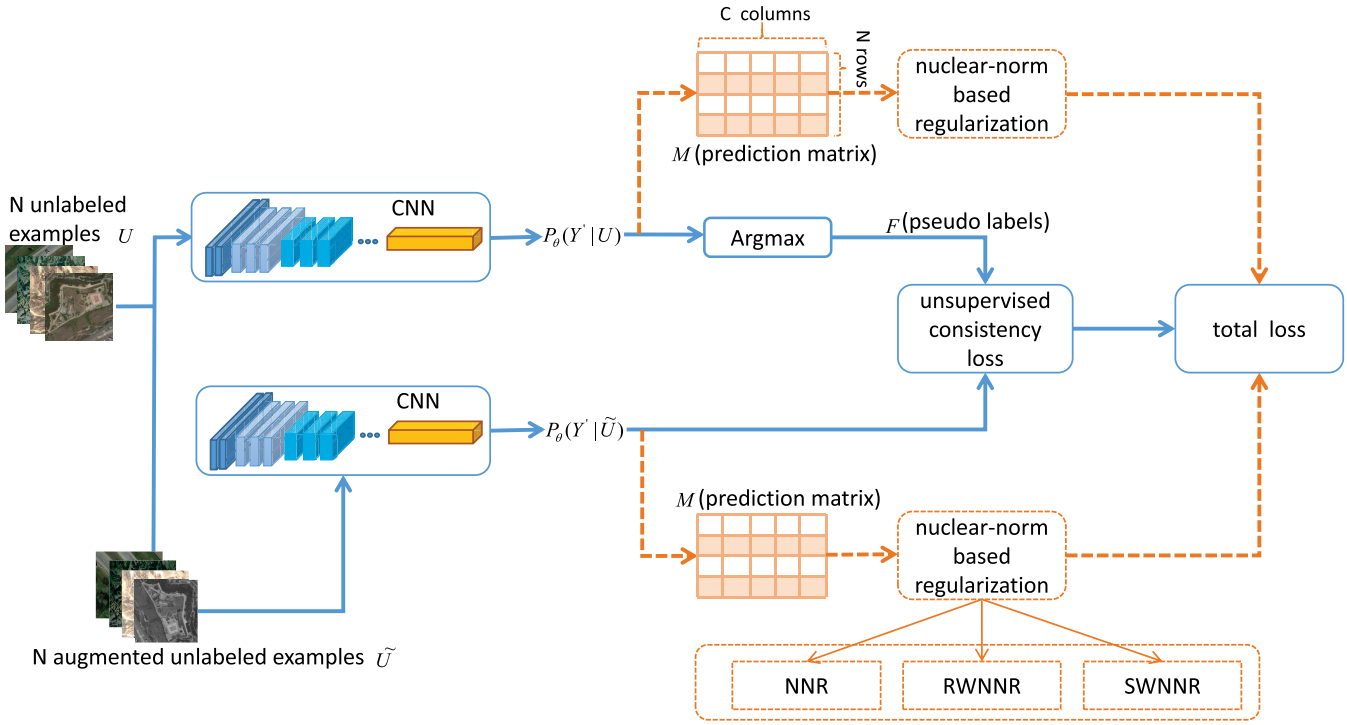


Fig. 2. **Schematic for the proposed WRMatch.** For the unlabeled examples U, the framework provides two branches, one of which predicts U and gets the pseudolabels F, and the other one predicts the perturbed $\tilde{U}$ and gets the probability distribution $P_\theta(Y'|X)$. The cross entropy between F and $P_\theta(Y'|X)$ becomes the consistency loss term. NNR, as shown in the orange dotted lines, regularizes the prediction matrix to improve the prediction diversity. In addition to NNR, we provide two kinds of weighting schemes for NNR to optimize the prediction discriminability: the RWNNR based on Dirichlet distribution and the SWNNR based on singular values themselves. For the labeled examples L, the framework treats them as common supervised learning, which is ignored in this figure for simplicity.

denotes a pseudolabel produced from the prediction. We refer to the cross entropy between two probability distributions p and q as $\mathrm{CE}(p, q)$.

FixMatch is a combination of two common SSL approaches: CR and pseudolabeling. For a batch of labeled examples L, FixMatch processes it as common supervised learning and calculates the standard cross entropy between X and Y as labeled loss term l_x [the definition is given in (1)]. As shown in Fig. 2, for a batch of unlabeled examples U, FixMatch provides two branches, one of which utilizes the CNN to predict U and converts the probability distributions into pseudolabels F, and the other one first augments U to get the disturbed examples $\tilde{U}$ and then uses the CNN to predict $\tilde{U}$ to get the probability distributions $P_\theta(Y'|\tilde{U})$. To obtain the pseudolabels for unlabeled examples, we first calculate the model's predicted class distributions, i.e., $P_\theta(Y'|U)$. Then, we use $\mathrm{argmax}(P_\theta(Y'|U))$ as the pseudolabels for the perturbed examples $\tilde{U}$. Finally, the cross entropy between the pseudolabels F and $P_\theta(Y'|\tilde{U})$ is calculated as the consistency loss term for the unlabeled data, which can be denoted as l_c. The loss terms l_x and l_c can be formulated as follows:

$$l_x = \frac{1}{N}\sum_{i=1}^{N} \mathrm{CE}\left(y_i, P_\theta\left(y_i'|x_i\right)\right) \tag{1}$$

$$l_c = \frac{1}{N}\sum_{i=1}^{N} I\left(\max\left(P_\theta\left(y_i'|u_i\right)\right) > \tau\right)\mathrm{CE}\left(f_i, P_\theta\left(y_i'|\tilde{u}_i\right)\right) \tag{2}$$

where $I(\cdot)$ is the indicator function and $I(\max(P_\theta(y_i'|u_i)) > \tau)$ can be considered as a confidence-based mask, which filters out predictions that the current model is not confident about.

The total loss function can be simply defined as follows:

$$\mathrm{loss} = l_x + l_c. \tag{3}$$

B. Diversity Collapse and Nuclear-Norm Regularization

However, as shown in Fig. 1, there are many similar categories in remote sensing scene images. The strong perturbation based on augmentation will lead to confusion between categories, e.g., the shapes of mountains and icebergs are very similar, but the only difference is the color of the ice and snow, which can be easily confused when color transformations are used as data perturbations. Due to the extreme lack of labels and strong perturbations, the model can easily generate confusing predictions, which can heavily decrease the prediction diversity.

To better observe prediction diversity intuitively, we compare the confusion matrix between FixMatch and supervised learning. As shown in Fig. 3, the experiments are carried out on the RESISC45 dataset with one label per category. After trained 25 000 iterations by the FixMatch, the model's prediction diversity declines seriously, i.e., most of the examples are recognized as *meadow*.

To address the diversity collapse problems, we first analyze the mathematical matrix corresponding to prediction (prediction matrix for short). Suppose that there are C categories

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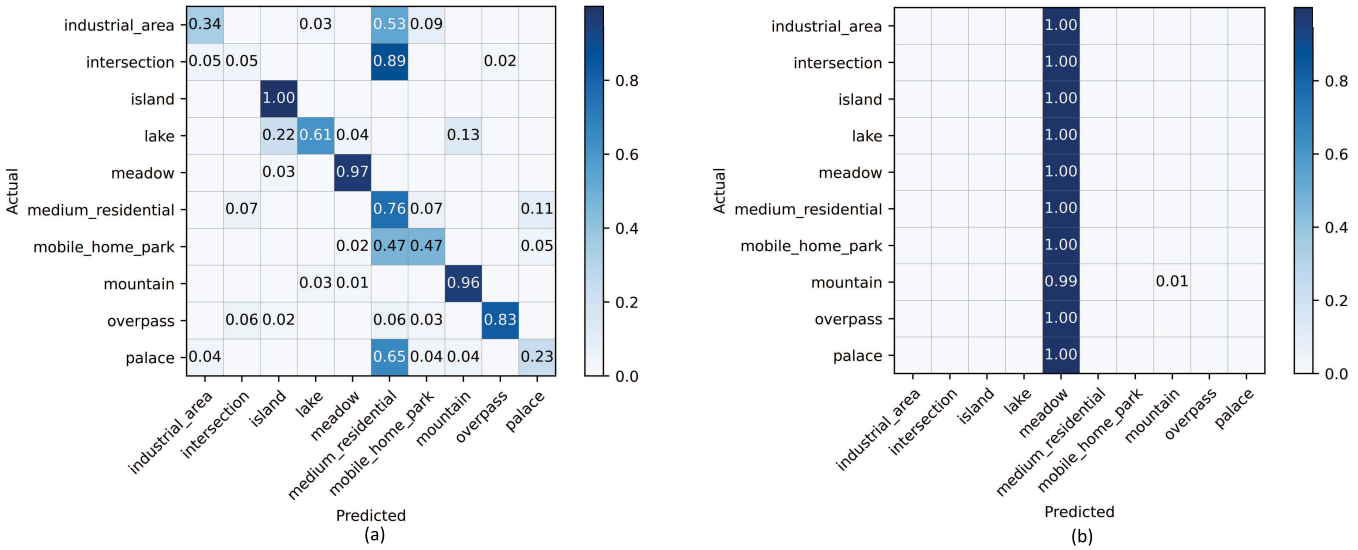


Fig. 3. **Confusion matrix of FixMatch and supervised learning.** For brief, we select ten categories from the original confusion matrix to display. Note that, since this figure is an intercept of the original confusion matrix and each value is rounded, the sum of each row may not equal 1. (a) Confusion matrix of supervised learning. (b) Confusion matrix of the FixMatch.

in the dataset, and the batch size is set to N during the training. In each iteration, the input of the neural network is a minibatch composed of N randomly selected examples, and the final output is an $N \times C$ matrix, denoted as M. Each row of M represents the prediction of an example, and each column represents the probability that the example belongs to a certain category. Because each row of the matrix represents the probability distribution of an example's prediction, higher diversity means greater differences between the rows of the matrix M. Hence, the prediction diversity can be measured by the matrix rank rank(M), which can represent the irrelevance between rows. Accordingly, we could maximize rank(M) to maintain prediction diversity.

Mathematically, the calculation of matrix rank is usually replaced by the nuclear-norm, which is the convex envelope of rank(M) [45]. Therefore, the prediction diversity could be maintained by maximizing $\|M\|_*$ approximately. The nuclear-norm of matrix M is defined as

$$\|M\|_* = \mathrm{tr}\left(\sqrt{M^T M}\right) = \sum_{i=1}^{D} \sigma_i \tag{4}$$

where tr(.) refers to the operation of calculating the trace of a matrix, $D = \min(N, C)$, and σ_i refers to the ith largest singular value of M.

Based on the above findings, we exploit the nuclear-norm to improve the prediction diversity. Specifically, we take the reciprocal of the nuclear-norm as the regularization term [denoted as nuclear-norm regularization (NNR)], such that minimizing the regularization term is maximizing the nuclear-norm. The regularization term is then applied to the prediction matrix of unlabeled data and that of the augmented unlabeled data and becomes two loss subterms l_u and $l_{\tilde{u}}$, which could be formulated as follows:

$$l_u = \frac{N}{\|P_\theta(\mathcal{U})\|_*} \tag{5}$$

$$l_{\tilde{u}} = \frac{N}{\|P_\theta(\tilde{\mathcal{U}})\|_*}. \tag{6}$$

The total loss is

$$l_{\text{total}} = l_x + l_c + \lambda(l_u + l_{\tilde{u}}) \tag{7}$$

where λ is a coefficient to adjust the weight of NNR in the total loss. For simplicity, l_u and $l_{\tilde{u}}$ share the same weight parameter λ.

C. Excessive Discriminability and Nuclear-Norm Weighting

NNR has two sides. Maximizing nuclear-norm not only improves prediction diversity but also indirectly affects the prediction discriminability because of the relationship among nuclear-norm, Frobenius-norm (F-norm), and the Shannon entropy [46]. Intrinsically, nuclear-norm and Frobenius-norm (F-norm) are the boundaries of each other. F-norm can be formulated as follows:

$$\|M\|_F = \sqrt{\sum_{i=1}^{N}\sum_{j=1}^{C} |M_{i,j}|^2}. \tag{8}$$

According to Recht *et al.* [47] and Srebro *et al.* [48], the mutual boundary relationships between $\|M\|_F$ and $\|M\|_*$ could be expressed as follows:

$$\frac{1}{\sqrt{D}}\|M\|_* \le \|M\|_F \le \|M\|_* \le \sqrt{D}\|M\|_F \tag{9}$$

where $D = \min(N, C)$.

Furthermore, Cui *et al.* [49] proved that $\|M\|_F$ is strictly opposite to the Shannon entropy in monotony, which can represent the prediction discriminability directly. Moreover, the minimum of the Shannon entropy and the maximum of the Frobenius-norm achieve at the same value When only one entry is 1 and other $C - 1$ entries are 0 in each row of M, under this condition, each example in the minibatch belongs

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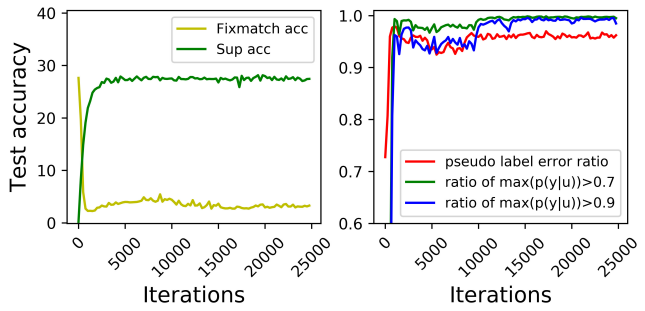


Fig. 4. **Overfitting phenomenon, pseudolabel error ratio, and ratio of $max(p(y|u))$ surpass the thresholds.** In the left subgraph, "Fixmatch acc" refers to the test accuracy of FixMatch, and "Sup acc" refers to the test accuracy of supervised learning. The right subgraph includes three training indicators of FixMatch.

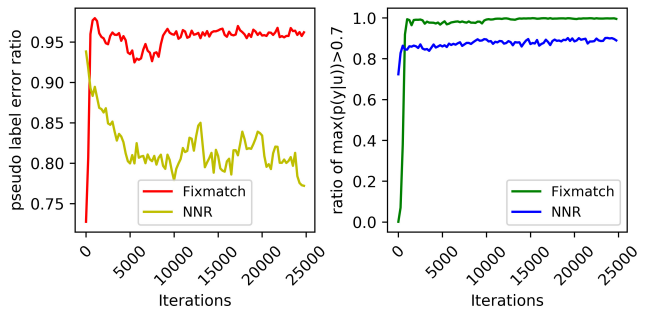


Fig. 5. Pseudolabel error ratio and ratio of $\max(p(y|u)) > 0.7$ after adopting NNR.

to a certain category, and the probability of belonging to other categories is zero, so the prediction has the highest discriminability, and the Shannon entropy achieves minimum 0. As a result, the nuclear-norm of the prediction matrix is relative to the prediction discriminability. Maximizing $\|M\|_*$ means high prediction confidence to some extent. In other words, maximizing the $\|M\|_*$ operation is in favor of prediction diversity but may not necessarily be a good solution taking discriminability into account.

As shown in the left subgraph of Fig. 4, when FixMatch is applied to the scenario with extreme lack of labels, a serious overfitting phenomenon appears, and the test accuracy drops dramatically. In the right subgraph, we can see that about 97% of the prediction values surpass 0.9, and about 98% of the prediction values surpass 0.7, which is overconfident and leads to a large number of error pseudolabels.

As shown in Fig. 5, after adopting NNR, although the discriminability of the prediction is somewhat reduced, the decrease is not significant due to the boundary relationship between the nuclear-norm and the Shannon entropy. Prediction is still overconfident (almost 90% of the prediction greater than 0.7), resulting in pseudolabels with an error rate of more than 75%.

As Gu *et al.* [50] pointed out, the standard nuclear-norm regularizes each singular value by the same weight factor, which is not the best scheme in many practical applications because singular values have different meanings. We would better treat each singular value with different weights according to specific scenarios. The weighted nuclear-norm can be formulated as follows:

$$\|M\|_{w*} = \sum_{i=1}^{D} w_i \sigma_i \tag{10}$$

where $w_i \geq 0$ is a nonnegative weight for σ_i.

Based on the aforementioned findings, we propose a novelty algorithm: WRMatch. We aim to maximize the weighted nuclear-norm with proposed weighting schemes to achieve two purposes concurrently: 1) improving the prediction diversity and 2) optimizing the prediction discriminability moderately. In the practical implementation, however, it is hard to design the weight for so many singular values manually. In this article, we propose two kinds of weighting algorithms: the simple RWNNR and the fine-grained SWNNR. In general, when the data characteristics are not fully understood and resources are not available for hyperparameter search, we can adopt the RWNNR to improve the generalization performance of the model to a certain extent. When there are enough computing resources, we can use the SWNNR to get better performance through hyperparameter search and meticulous hyperparameter setting.

D. Random Weighted Nuclear-Norm Regularization

We, first, design a weighting scheme from the perspective of model generalization. According to the relationship between the nuclear-norm and the Shannon entropy, randomly increasing or decreasing the weight of singular value can lead to the change of the discriminability of prediction accordingly. To minimize the loss, the optimizer continuously optimizes the model and outputs predictions that are more consistent with the objective in the training process. When the model converges, the prediction discriminability will reach a suitable interval, in which the model can classify scenes correctly even if the prediction discriminability is randomly increased or decreased within a certain range.

Here, we subject the weighting coefficients $\boldsymbol{a} = \{a_1, a_2, \ldots, a_D\}$ to the Dirichlet distribution: $\boldsymbol{a} \sim \text{Dir}(\boldsymbol{\beta})$, where $\boldsymbol{\beta} = \{\beta_1, \beta_2, \ldots, \beta_D\}$ is the concentration parameter. The probability density function of the Dirichlet distribution is defined as

$$\text{Dir}(\boldsymbol{a}|\boldsymbol{\beta}) = \frac{1}{B(\beta)} \prod_{i=1}^{D} a_i^{\beta_i - 1} \tag{11}$$

$$B(\boldsymbol{a}) = \frac{\prod_{i=1}^{D} \Gamma(a_i)}{\Gamma(a_0)}, \quad a_0 = \sum_{i=1}^{D} a_i, \quad D \geq 3 \tag{12}$$

where $\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$ is the gamma function.

In the standard nuclear-norm, each singular value has the same weight. Therefore, we adopt the symmetric Dirichlet distribution, which is a special case of the Dirichlet distribution, and the concentration parameter is the same in all D dimensions. For simplify, we set $\boldsymbol{\beta} = \{1, 1, \ldots, 1\}$. In the training phase, we sample a group of weight $\boldsymbol{a}$ for singular

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values in each iteration. Formula (10) can be replaced by

$$\|M\|_{s*} = D\sum_{i=1}^{D} a_i\sigma_i. \tag{13}$$

Because $\sum_{i=1}^{D} a_i = 1$ when $\boldsymbol{a} \sim \mathrm{Dir}(\boldsymbol{\beta})$, while the sum of weights in the standard nuclear-norm equals D, so we should multiply the random weighted nuclear-norm by D. Accordingly, formulas (5) and (6) are reformulated as follows:

$$l_{us} = \frac{N}{\|P_\theta(\mathcal{U})\|_{s*}} \tag{14}$$

$$l_{\tilde{u}s} = \frac{N}{\|P_\theta(\tilde{\mathcal{U}})\|_{s*}}. \tag{15}$$

Through the random perturbation of singular values, the weighted regularization term enables the model to adapt to the variation in features, which indirectly enhances the model's adaptability to the variation in input images. Therefore, the model achieves a better generalization ability.

E. Self-Weighted Nuclear-Norm Regularization

Although random weighting can improve the model generalization, this method lacks the ability of fine-grained setting according to the difficulty of the issue. In this section, we explore a novelty fine-grained weighting scheme.

Large singular values reflect the primary information of the prediction matrix, while small singular values reflect the subtle information. In other words, large singular values represent large prediction values in the matrix (maybe overconfident predictions). Thus, when maximizing $\|M\|_*$, we try to moderately reduce the weight of large singular values and increase the weight of small singular values.

It is difficult to determine each singular value's best coefficient, but there is a basic principle: the larger the singular value, the smaller the coefficient. Therefore, we propose an effective weighting method according to singular values themselves

$$\|M\|_{k*} = \left(\sum_{i=1}^{D} {\sigma_i}^{1/k}\right)^k \tag{16}$$

where $k = \{1, 2, 4, \ldots, 2^\upsilon\}$ is an adjustment factor and υ is a nonnegative integer. In this formula, $\sigma_i^{(1-k)/k}$ is treated as the weight for σ_i. We can see that the bigger σ_i is, the smaller the weight. That is to say, the weight is adaptive to the singular values themselves. Moreover, the bigger k is, the more ${\sigma_i}^{1/k}$ approaches 1, and the more severe the inhibition on the large singular value is.

For $k = 2$, we can prove that $\|M\|_{2*}$ and $\|M\|_*$ are the boundaries of each other

$$\|M\|_{2*} = \left(\sum_{i=1}^{D} {\sigma_i}^{1/2}\right)^2 \le D\sum_{i=1}^{D}\sigma_i = D\|M\|_* \tag{17}$$

$$\|M\|_{2*} = \left(\sum_{i=1}^{D} {\sigma_i}^{1/2}\right)^2 \ge \sum_{i=1}^{D}\sigma_i = \|M\|_*. \tag{18}$$

Combining (17) and (18), we can get

$$\frac{1}{D}\|M\|_{2*} \le \|M\|_* \le \|M\|_{2*} \le D\|M\|_*. \tag{19}$$

Then, combining (19) and (9), we infer that $\|M\|_{2*}$ and $\|M\|_F$ are also the boundaries of each other

$$\frac{1}{D\sqrt{D}}\|M\|_{2*} \le \|M\|_F \le \|M\|_{2*} \le D\sqrt{D}\|M\|_F. \tag{20}$$

Note that the boundaries become more relaxed, which means that the discriminability could vary in a larger range when maximizing $\|M\|_{2*}$.

In the same way, we can continuously prove that $\|M\|_{k*}$ and $\|M\|_F$ are the boundaries of each other, with more and more relaxed constraints. Based on this SWNNR, we replace formulas (5) and (6) as follows:

$$l_{uk} = \frac{N}{\|P_\theta(\mathcal{U})\|_{k*}} \tag{21}$$

$$l_{\tilde{u}k} = \frac{N}{\|P_\theta(\tilde{\mathcal{U}})\|_{k*}}. \tag{22}$$

In this loss function, each singular value is weighted adaptively according to the singular value itself. In addition, the degree of weighting can be further optimized by the hyperparameter k, which should be fine-grained adjusted according to the prediction discriminability.

IV. Experiments

In this section, we evaluate our proposed WRMatch in three public datasets for remote sensing scene classification. To demonstrate the superiority of our method, we report the experimental results and compare WRMatch with other SOTA methods. We conduct a set of ablation experiments to determine how each factor affects the model performance and further validate how our approach can improve prediction diversity and optimize prediction discriminability. Besides, we also observe and analyze the influence of the self-weighting factor k on the model performance.

A. Dataset Description

In our experiments, we have evaluated our proposed method on three public datasets for the remote sensing scene classification task, including the UC Merced dataset [51], the AID dataset [52], and the RESISC45 dataset [27].

The UC Merced dataset [51] is a standard benchmark for remote sensing scene classification tasks, and it consists of 21 categories of scene images for land-use research. For each scene class, there are 100 aerial images with a spatial resolution of 1 ft. The original images were downloaded from the United States Geological Survey (USGS) National Map of 20 regions and then were cut into the size of 256 × 256 pixels.

The Aerial Image dataset (AID) [52] is a large-scale dataset for aerial scene classification by collecting sample images from Google Earth imagery, which contains 30 scene categories with a total of 10 000 images (each category has about 220∼420 images). The pixel resolution changes from about 8 to about 0.5 m, and each image is fixed to a uniform size of 600 × 600 pixels. The images in AID are multisource, taken from different sensors and different countries and regions, making it more challenging.

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TABLE I
COMPARISON BETWEEN DIFFERENT NUMBERS OF LABELS ON ALL THREE DATASETS

Number of labels	WRMatch (SWNNR)			WRMatch (RWNNR)		
	RESISC45	AID	UC Merced	RESISC45	AID	UC Merced
L=1	31.41±2.08	41.63±2.17	39.52±3.14	29.35±2.12	40.68±1.69	40.19±2.47
L=2	62.87±1.81	70.78±1.76	64.76±1.79	61.45±1.71	72.65±2.85	62.38±2.64
L=5	83.79±0.79	89.93±0.97	85.51±1.76	83.17±0.73	87.85±1.18	81.25±1.12
L=10	**89.03 ±0.56**	**94.80±0.54**	**94.76±0.67**	**88.76±0.57**	**93.84±0.73**	**89.29±0.75**

The RESISC45 dataset [27] is another challenging dataset for remote sensing scene classification, created by Northwestern Polytechnical University (NWPU). The dataset consists of 45 categories with 700 images in each category, totaling 31 500 images. The resolution varies from 30 to 0.2 m per pixel for most images, and the size of each image is 256 × 256 pixels. There are three remarkable characteristics for this dataset: first, large scale and various categories; second, rich variations in illumination, perspective, background, and so on; and third, high interclass similarity and intraclass diversity.

B. Experiment Settings

In the practical implementation, each dataset is divided randomly into three parts: 60% as the training set and 20% as the validation set while the remaining 20% as the test set. Then, we randomly select a few (one, two, five, or ten) examples per category from the training set as the labeled subset and treat the remaining of the training set as the unlabeled subset. In Fig. 2, two CNNs adopt the same neural network architecture and share the parameters. We choose ResNext50 [53] as the backbone architecture for all the experiments unless otherwise specified and initialize the model with parameters pretrained on ImageNet. During the feature extraction processing, all the images were resized to 224×224 to meet the size of CNN's input. We utilize SGD as the basic optimizer and set the weight decay to 0.0005 with the momentum to 0.9; also, we adopt stochastic weight averaging [54] to improve the model generalization. The learning rate is set to 0.001 with the warm-up strategy, while the batch size N is set to 16. For the RESISC45 and AID datasets, we train the model with 25 000 iterations and evaluate the model every 250 iterations. For the relatively small and easy dataset, UC Merced, we train the model with 5000 iterations and evaluate the model every 100 iterations. For the coefficient λ in the total loss function and the threshold τ for the pseudolabel, we find that setting λ to 0.25 and τ to 0.7 could provide satisfying performance. The parameter k in the self-weighted NNR (WNNR) needs to be adjusted according to the complexity of the dataset and the number of labels: for RESIC45, we set $k = 8$ when one or two labels per category and $k = 2$ when five or ten labels per category; for AID, we set $k = 2$ when one, two, or five labels per category and $k = 1$ when ten labels per category; and for UC Merced, we set $k = 1$ always. Our model has 23.07M learnable parameters, and the calculation load is 4.26 GFLOPs. In the practical implementation, we train the model with the PyTorch framework and rely on a single GPU (2080Ti) as the hardware platform. It takes about 4 h to train for the RESISC45 and AID datasets, respectively, and about 1 h for the training of the UC Merced dataset. In the inference phase, each image takes 0.03 s. The general evaluation criteria in this article include the overall accuracy (OA) and the confusion matrix. The mean and standard deviation of OA values are calculated after five runs.

C. Experimental Results and Comparisons

1) Results Obtained Using Different Number of Labels on Three Datasets: To investigate the algorithm behaviors under different labeled data scales, we first make a quantitative comparison on three datasets. We report the accuracy results of utilizing different numbers of labels per category, as shown in Table I. It can be seen that WNNR is very conducive to the model's generalization so that the model can correctly predict when extreme lack of labels, without falling into diversity collapse or overfitting. In general, with fine-grained settings, SWNNR achieves better testing accuracy than RWNNR. Besides, we also report extra accuracy results with Resnet34 and VGG16 as backbones, as shown in Tables II and III. It can be seen that all three backbones get superior performance, and Resnext50 is better than the other two. Intuitively, we choose a representative metalearning method MAML, a classical transfer learning method with GoogleNet, and our WRMatch to recognize images from all three datasets, as shown in Fig. 6. The models are all trained with five labels per category. It can be seen that WRMatch has the best prediction accuracy. Moreover, when the image has simple or regular structures (such as *Meadow* and *Airport*), MAML performs better than transfer learning, and when the image structures are complex or the edge details are abundant (such as *Church* and *Island*), the performance of transfer learning is better than that of MAML. That is because MAML has better generalization due to its large number of metatasks when images have simple or regular structures, while transfer learning inherits the ability to extract details learned from Imagenet.

2) Comparison With Few-Shot Learning Methods: Recently, some efforts have been made to utilize few-shot learning for remote sensing scene classification, which only needs a few labels to recognize the unseen category. In this part, we aim to compare the proposed method to widely used few-shot learning methods. As shown in Tables IV–VI, our method surpasses the most metalearning approaches and is a fairly competitive solution. Specifically, when there is only one label per category, WRMatch performs better than general transfer learning but worse than advanced metalearning

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TABLE II

COMPARISON BETWEEN DIFFERENT NUMBERS OF LABELS ON ALL THREE DATASETS (WITH BACKBONE RESNET34)

Number of labels	WRMatch (SWNNR)			WRMatch (RWNNR)		
	RESISC45	AID	UC Merced	RESISC45	AID	UC Merced
L=1	30.25±1.83	45.81±1.40	38.73±3.60	29.59±2.07	37.38±2.14	30.24±1.36
L=2	50.81±1.89	66.53±1.93	59.76±2.04	52.05±2.01	60.41±2.21	61.50±2.04
L=5	76.00±0.50	87.36±0.80	81.75±1.25	74.13±0.59	86.95±0.59	80.31±0.63
L=10	**82.79 ±0.67**	**93.73±0.13**	**88.73±0.49**	**82.78±0.65**	**93.26±0.27**	**87.78±1.05**

TABLE III

COMPARISON BETWEEN DIFFERENT NUMBERS OF LABELS ON ALL THREE DATASETS (WITH BACKBONE VGG16)

Number of labels	WRMatch (SWNNR)			WRMatch (RWNNR)		
	RESISC45	AID	UC Merced	RESISC45	AID	UC Merced
L=1	36.70±1.83	51.85±2.20	43.72±2.88	31.04±1.57	42.38±3.12	46.14±2.52
L=2	62.14±2.01	61.51±2.41	66.58±1.76	52.30±0.67	62.41±2.38	71.75±1.74
L=5	76.21±0.51	83.01±1.25	82.54±1.17	74.40±0.69	85.08±0.82	81.58±1.46
L=10	**82.92 ±0.42**	**92.01±0.27**	**88.25±0.64**	**82.23±0.18**	**91.51±0.32**	**88.13±0.79**

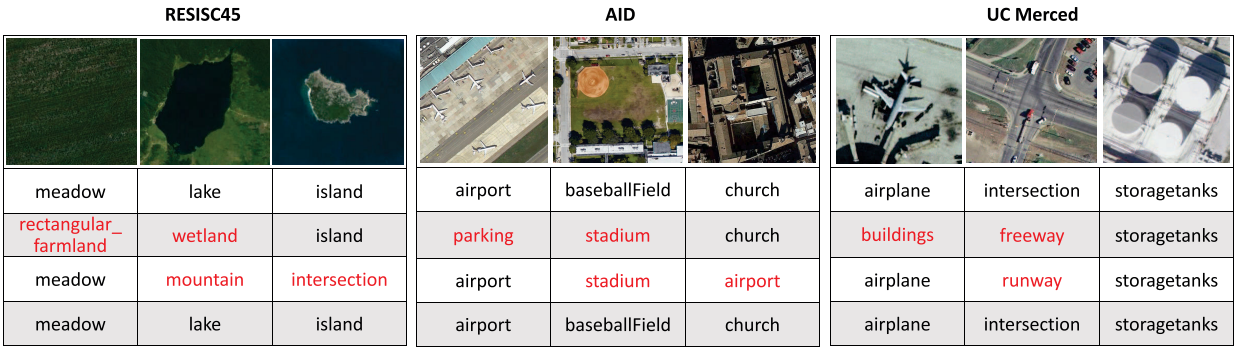


Fig. 6. **Prediction results of WRMatch, transfer learning with GoogleNet, and metalearning method MAML on (Left) RESISC45, (Middle) AID, and (Right) UC Merced.** The wrong predictions are highlighted using the red font.

TABLE IV

PERFORMANCE COMPARISON BETWEEN WRMATCH'S WITH OTHER METHODS ON THE RESISC45 DATASET

Methods	Number of labels		
	L=1	L=5	L=10
transfer with GoogleNet [1]	25.91±1.42	37.66±1.32	55.35±1.39
transfer with ResNet152 [1]	20.71±1.80	27.18±1.56	35.18±1.99
PrototypicalNet [24], [2]	40.33±0.18	63.82±0.56	-
RelationNet [55], [2]	66.43±0.73	78.35±0.51	-
MAML [1]	42.29±0.76	61.85±0.81	68.77±0.69
RS-MetaNet [1]	52.78±0.09	71.49±0.81	77.37±0.77
DLA-MatchNet [2]	**68.80±0.70**	81.63±0.46	-
WRMatch (RWNNR)	29.35±2.12	**83.17±0.73**	**88.76±0.57**
WRMatch (SWNNR)	31.41±2.08	**83.79±0.79**	**89.03±0.56**

TABLE V

PERFORMANCE COMPARISON BETWEEN WRMATCH'S WITH OTHER METHODS ON THE AID DATASET

Methods	Number of labels		
	L=1	L=5	L=10
transfer with GoogleNet [1]	20.76±1.29	40.67±1.18	55.63±1.35
transfer with ResNet152 [1]	20.67±1.09	23.66±1.30	34.90±1.62
PrototypicalNet [24], [26]	61.49	81.97	85.38
RelationNet [55], [26]	65.99	77.04	78.90
MAML [1]	47.93±0.72	61.79±0.75	69.90±0.70
RS-MetaNet [1]	**56.32±0.55**	74.48±1.11	80.57±0.61
WRMatch (RWNNR)	40.68±1.69	**87.85±1.18**	**93.84±0.73**
WRMatch (SWNNR)	41.63±2.17	**89.93±0.97**	**94.80±0.54**

methods. When there are more than five tags per category, WRMatch shows a huge advantage over other SOTA methods by a large margin. In the one-shot scenario, there are two reasons why WRMatch is inferior to metalearning. First, the metalearning adopts a large number of remote sensing images in the metatraining stage, making the model adapt to the distribution of remote sensing images, so the model has better parameter initialization compared with WRMatch in this case. Second, metalearning builds a large number of tasks, and each task contains different categories and examples, thus improving the generalization of the model. Therefore, when encountering the unseen categories, metalearning can still recognize them well even if there is only one label for

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TABLE VI
PERFORMANCE COMPARISON BETWEEN WRMATCH'S WITH OTHER METHODS ON THE UC MERCED DATASET

Methods	Number of labels		
	L=1	L=5	L=10
transfer with GoogleNet [1]	23.45±1.29	45.22±1.48	55.59±0.94
transfer with ResNet152 [1]	20.72±0.41	29.29±0.41	31.61±1.55
PrototypicalNet [24], [26]	50.45	79.82	84.40
RelationNet [55], [26]	57.70	65.79	70.83
MAML [1]	47.53±0.71	63.13±0.92	64.99±0.91
RS-MetaNet [1]	**57.23±0.56**	76.08±0.28	81.23±0.45
DLA-MatchNet [2]	53.76±0.62	63.01±0.51	-
WRMatch (RWNNR)	40.19±2.47	**81.25±1.12**	**89.29±0.75**
WRMatch (SWNNR)	39.52±3.14	**85.51±1.76**	**94.76±0.67**

each category. With the increase in the number of labels, WRMatch dramatically improved the generalization ability due to the WNNR (for example, with two labels per class, the accuracy of WRMatch on RESISC45 dramatically increases to 62.87%). When there are five labels per class, the generalization ability of WRMatch greatly surpasses other metalearning methods. It should be pointed out that there is a metatraining phase in metalearning, which still needs a large number of labels. This method can only alleviate the need for labels of the unseen category during the metatest phase. In contrast, WRMatch does not have the metatraining stage nor does it need many labels in that stage. Therefore, WRMatch may be a better solution in some scenarios. For example, when the labels required for the metatraining phase are also scarce, metalearning is no longer available, but WRMatch is still applicable.

D. Ablation Study

As WRMatch comprises several improvements, we also carry out an extensive ablation study to tease apart the experimental factors that are most important to improve the classification performance. We randomly select a few labels from each category of the dataset and perform an ablation study to examine the FixMatch, NNR, random WNNR, and self-WNNR. The results are reported in Fig. 7, where WRMatch(R) refers to WRMatch with random WNNR and WRMatch(S) refers to WRMatch with self-WNNR.

When only one or two labels per category are used, FixMatch brings about a severe decline in accuracy. NNR overcomes the severe decline caused by FixMatch, and WRMatch further improves the accuracy. FixMatch suffers serious performance degradation due to the pseudolabeling, which improves the discriminability of the prediction and leads to serious overfitting. When there are more than five labels per category, FixMatch no longer brings overfitting and has a great performance improvement compared with the baseline methods. In this case, NNR can still improve the performance, while WRMatch can further improve the classification accuracy on the basis of NNR.

E. Improvements in Prediction Diversity

As shown in Fig. 8, we compare the confusion matrix on the RESISC45 dataset with one label per category. When

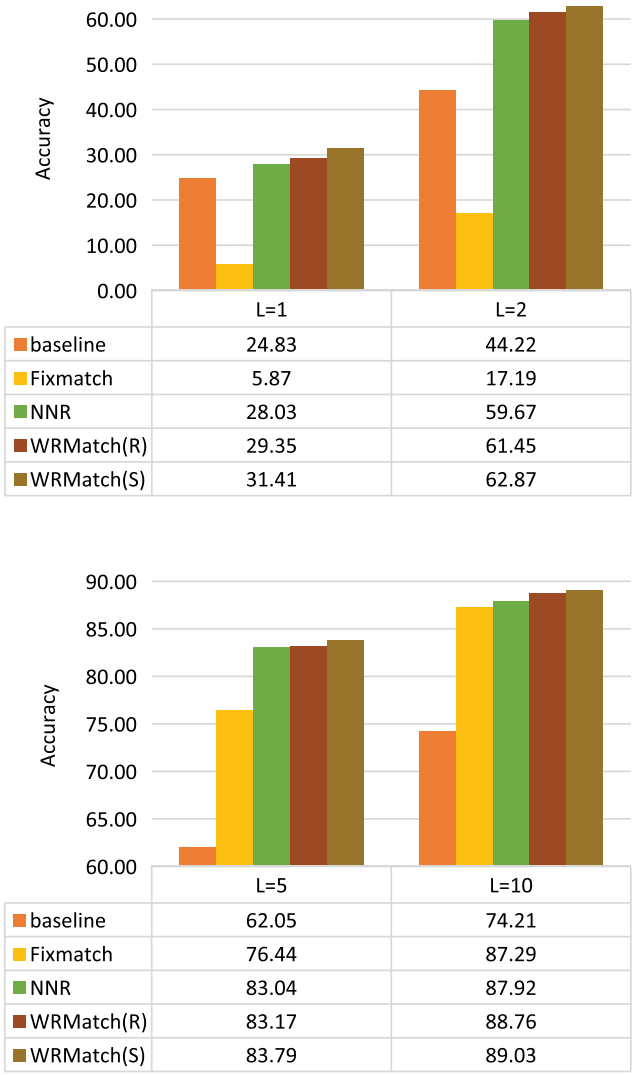


Fig. 7. Ablation study of WRMatch on the RESISC45 dataset.

there are more elements on the matrix's diagonal, the greater the element value, the better the diversity of the prediction is. By observing the confusion matrix (a) and (b), we can see that FixMatch brings about a severe decline in prediction diversity, but WNNR of WRMatch can recover and maintain the prediction diversity. For example, on the RESISC45 dataset, the supervised learning (as shown in Fig. 3) only mistakenly predicts a small amount of three categories: *lake, mobile_home_park*, and *mountain* as *meadow*, but FixMatch (a) predicts most of the categories as *meadow*, while WRMatch (b) only mistakenly predicts a small amount of *mobile_home_park* as *meadow*, which even outperforms the supervised learning. This means that WRMatch reduces confusion between categories and improves prediction diversity. Besides, we compare the confusion matrix among WRMatch (b), transfer learning with GoogleNet (c), and representative metalearning method MAML (d). It can be seen that the prediction diversity of WRMatch is better than that of transfer learning with GoogleNet, but MAML is even better than WRMatch because of its diverse metatasks.

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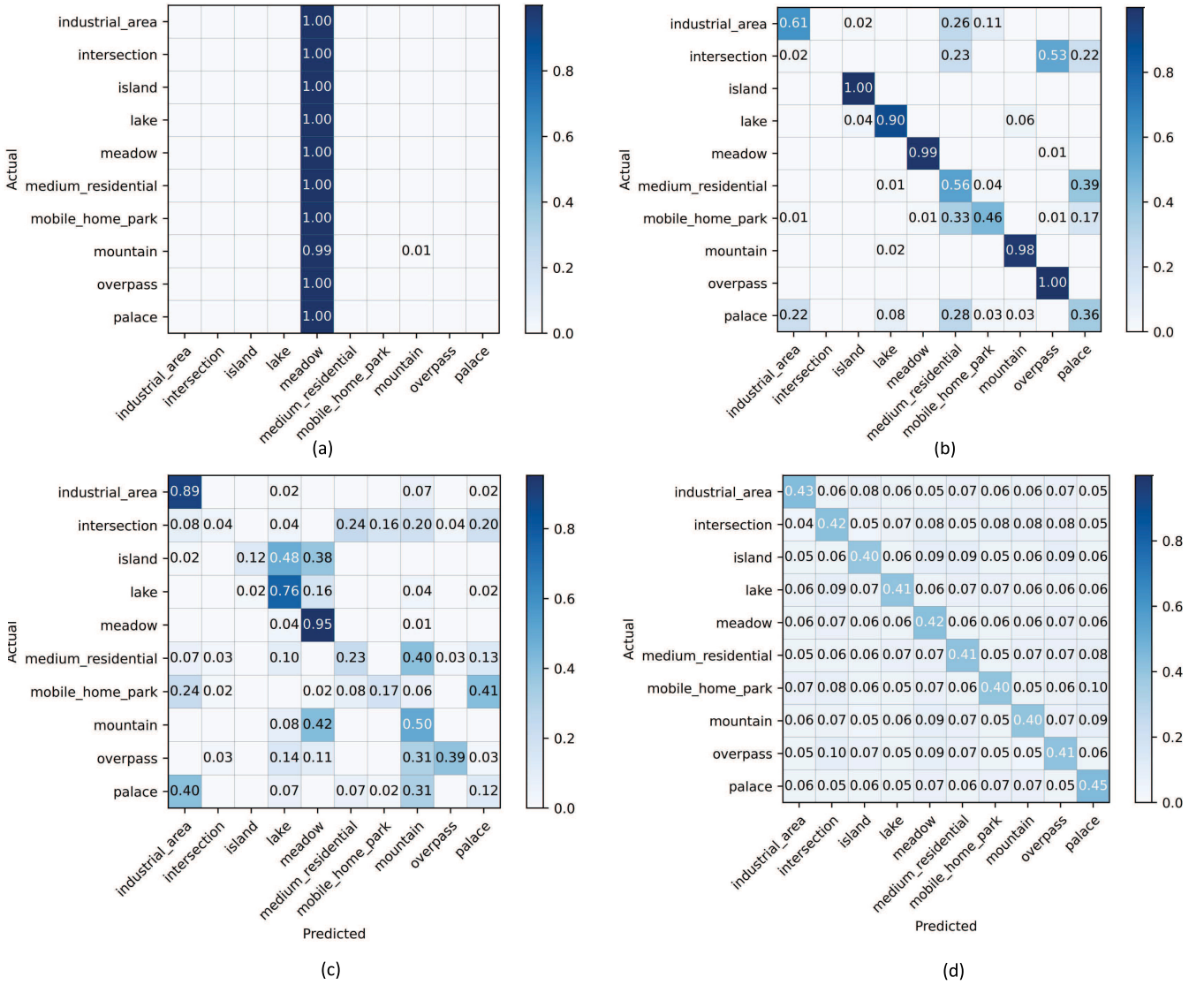


Fig. 8. **Confusion matrix of (a) FixMatch, (b) WRMatch, (c) transfer learning with GoogleNet, and (d) MAML.** For clarity, we select ten categories from the original confusion matrix to display. Note that, since this figure is an intercept of the original confusion matrix and each value is rounded, the sum of each row may not equal 1.

F. Optimization of Prediction Discriminability

To illustrate why WNNR performs well, we analyze the pseudolabel error ratio, and the ratio of $\max(p(y|u))$ surpasses the thresholds for FixMatch and WRMatch with SWNNR in Fig. 9. The experiments are carried out on the RESISC45 dataset with one label per category. By comparing the curves in the middle and right subgraphs, we can see that FixMatch has much higher prediction discriminability, while NNR degrades the prediction confidence. Moreover, SWNNR with $k = 8$ further decreases the confidence to a more suitable range because of the more relaxed constraint. As a result, NNR has a more low error ratio of the pseudolabel, and SWNNR with $k = 8$ further decreases the error ratio, which is shown in the left subgraph.

G. Analysis of the Self-Weighting Factor k

In WRMatch with self-WNNR, factor k can adjust the prediction discriminability and has an important effect on the prediction performance. We set $k = \{1, 2, 4, 8, 16\}$ and compare the prediction accuracy under different k values. As shown in Fig. 10, we have drawn the comparison curves with different labels, where L refers to the number of labels for each category. When $L = 1$ or $L = 2$, model achieves the highest accuracy with $k = 8$, and when $L = 5$ or $L = 10$, model achieves the highest accuracy with $k = 2$. We can see that, when there is an extreme lack of labels, the model is prone to overfitting, so a large k value is needed to reduce the prediction discriminability. When there are more labels, the model is no longer easy to overfitting, and the best k value can be relatively small. Besides, with a different number of labels, the degree of accuracy improvement brought by self-weighting is different. When $L = 1$, the appropriate k can improve the accuracy by 3.38%; When $L = 2$, the accuracy increased by 3.20%; when $L = 5$ and $L = 10$, the improvement of accuracy brought by self-weighting becomes small (respectively, 0.75% and 1.11%). Therefore, we can see that the fewer the labels, the more serious the

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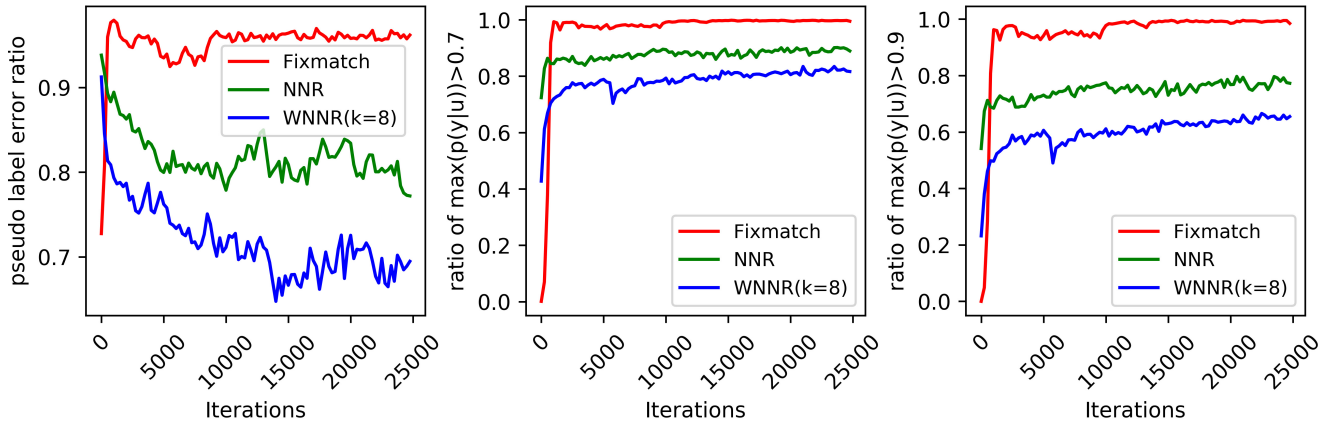


Fig. 9. **Pseudolabel error ratio and the ratio of $max(p(y|u))$ surpass the thresholds for FixMatch, NNR, and WNNR.** The curves reflect the moderate decrease in the prediction discriminability after adding SWNNR, which leads to the degradation of the pseudolabel error ratio.

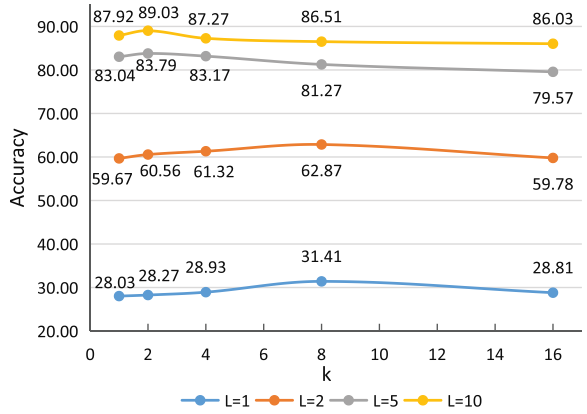


Fig. 10. Effect of the self-weighting factor k on accuracy.

overfitting, and the more prominent the effect of nuclear-norm weighting.

V. Discussion

A. Strengths

WRMatch provides an excellent SSL method for few-shot remote sensing scene classification, which achieves competitive performance and can be trained in an end-to-end manner. The weighted nuclear-norm can not only improve the prediction diversity but also optimize the prediction discriminability, thus avoiding the overfitting phenomenon and diversity collapse in the few-shot scenario. As shown in Tables IV–VI, WRMatch outperforms all other methods by a large margin on all three datasets when five or ten labels per category. Compared with metalearning, WRMatch is trained in an end-to-end way, and there is no metatraining phase composed of multiple subtasks, which requires additional categories with lots of labels. WRMatch provides an effective and competitive method for few-shot scene classification in remote sensing, especially suitable for the situation that there is no additional category with many labels, but there are lots of unlabeled examples.

B. Limitations

Although the model is initialized with Imagenet pretrained parameters, there exists a large domain gap between natural images and remote sensing images, so the initialization condition of WRMatch should be further improved. On the contrary, in the metatraining stage, metalearning utilizes a large number of remote sensing images of other categories as prior knowledge. Thus, when only one label per category is used, the accuracy of WRMatch is worse than that of the metalearning methods. Besides, the self-WNNR needs more hyper-parameters, which should be adjusted according to the complexity of the dataset and the number of labels. Future work will explore how to adjust the k value automatically for a more intelligent self-weighting algorithm.

C. Combination of Semisupervised Learning and Metalearning

SSL and metalearning have their advantages and are suitable for different scenarios. How to combine the two is meaningful research in the future. Although there have been some attempts to promote the generalization of metalearning by using unlabeled samples, these studies are not sufficient, and there is still room for improvement. On the other hand, in SSL, some categories are easily confused with others due to the lack of labels. Drawing the idea of metalearning, we can divide the classification tasks in semisupervision into several subtasks with a small number of categories, which may be a feasible way to alleviate confusion.

VI. Conclusion

In this article, we propose an improving semisupervised algorithm, WRMatch, to alleviate the label requirement for remote sensing scene classification. WRMatch utilizes WNNR to improve the prediction diversity and optimize the prediction discriminability simultaneously. We provide two weighting schemes to weight the singular values: the RWNNR based on Dirichlet distribution and the SWNNR to weight the singular values according to singular values themselves. Comparison

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experiments with the other SOTA methods show the superiority of WRMatch, and extensive ablation studies are conducted to tease apart the experimental factors that are important to improve the classification performance. The experimental results show that our method provides superior performance for the few-shot scene classification in remote sensing.

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